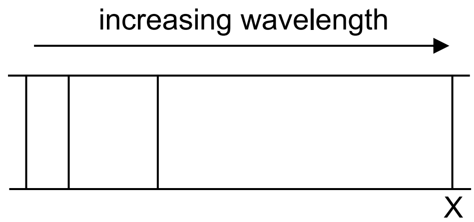
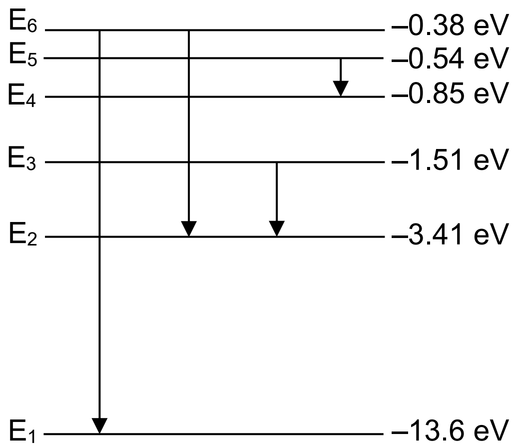
30 The line emission spectrum of a hydrogen discharge tube when viewed through a diffraction grating consists of coloured lines on a dark background. One of the coloured lines is marked X, as shown.

The wavelength scale is linear and increases to the right.



The diagram below represents some of the electron energy levels in a hydrogen atom.



Which transition best corresponds to the emission of line X?

A E_6 to E_1 **B** E_6 to E_2 **C** E_3 to E_2 **D** E_5 to E_4

Ans: C

Since the line spectrum consists of coloured lines on dark background, the spectrum must fall within the visible light wavelengths of about 400nm to 700nm (1.8 eV to 3.1 eV). Since $\Delta E = \frac{hc}{\lambda}$, only transitions E_6 to E_2 and E_3 to E_2 will emit photons of wavelengths within the visible spectrum. Since the visible emission line X has the longest visible wavelength, it must have the smallest energy difference between the levels, thus E_3 to E_2. Transition E_6 to E_1 falls within the UV spectrum whereas transition E_5 to E_4 falls within the infrared spectrum.

End of Paper